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/ **2026 AHFTC Board Certification Question Bank**
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/ **Practice Exam - 1**

Correct Answer

Next Question

Practice Exam - 1

Question 7 of 117



A 70 year-old female with a history of hypertension, obesity (BMI 38 Kg/m²), and prior smoking (20 pack-years) presents for evaluation of dyspnea. For the last year, she has noted dyspnea and associated chest tightness ascending one flight of stairs or walking up inclines. Her heart rate is 82 bpm and blood pressure 131/83 mmHg. She takes amlodipine 10 mg daily. JVP is difficult to assess. The heart rate is regular with distant S1 and S2 and no murmurs or gallops. The lung fields are without crackles or wheezes. There is trace pedal edema. ECG demonstrates sinus rhythm. Chest x-ray is unremarkable. Echocardiogram shows normal left ventricular wall thickness, chamber size, and ejection fraction of 55-60%. She undergoes cardiac catheterization, showing no coronary artery disease.

Which of the following additional findings would be most consistent with a diagnosis of heart failure with preserved ejection fraction?

A. E/e' ratio of 10



B. NT-proBNP of 120 pg/ml

C. Left atrial volume index of 30 ml/m²



D. Pulmonary capillary wedge pressure of 16 mmHg

Commentary

References

Testing Point: The Board-certified AHFTC specialist should be able to diagnose heart failure with preserved ejection fraction.

Correct Answer: D. Pulmonary capillary wedge pressure of 16 mmHg.

Rationale: A diagnosis of HFpEF requires clinical signs/symptoms of heart failure, evidence of normal/preserved ejection fraction, and echocardiographic or hemodynamic evidence of elevated left ventricular filling pressures. This patient has dyspnea and preserved LVEF and the risk factors of age and obesity. A pulmonary capillary wedge pressure of > 15 mmHg at rest is abnormal and definitive for a diagnosis of HFpEF in this scenario.

Answer A: The E/e' ratio from Doppler echocardiography can provide a non-invasive estimation of left ventricular diastolic function

that correlates with invasive measurement of left ventricular filling pressures. There is consensus that E/e' ratio >15 is most supportive of the diagnosis of HFpEF, with values 10-14 having less strong association.

Answer B: Elevated B-type natriuretic peptide levels are useful in the diagnosis of HFpEF, but normal values do not exclude heart failure. Levels may be normal particularly in obese patients with heart failure. NT-proBNP level >220 pg/ mL is supportive of a HFpEF diagnosis. The NT-proBNP value in this patient is not supportive, so other evidence is needed.

Answer C: Left atrial enlargement, consistent with increased left ventricular filling pressures, can be useful in the diagnosis of HFpEF. There is consensus that left atrial volume index > 34 mL/m² supports a diagnosis of HFpEF. The left atrial volume index of this patient is within normal limits.

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